3.22 Lying over, Incomparability, Going Up and Going Down

Definition 3.22.1 (Lying over / Going up)
Let $\phi : A \to B$ be a ring map and $\mathfrak{p}$ and $\mathfrak{q}$ primes of A and B respectively

a) *$\mathfrak{q}$ **lies over** $\mathfrak{p}$, or $\mathfrak{p}$ **lies under** $\mathfrak{q}$ if $\mathfrak{p} = \phi^{-1}(\mathfrak{q}) = \mathfrak{q}^c$. When $A \subseteq B$ and ϕ is the identity then this is equivalent to saying $\mathfrak{p} = \mathfrak{q} \cap A$.*

Definition 3.22.2 (Lying Over / Going Up / Incomparability)
Let $\phi : A \to B$ be a ring map. We say that it has the

a) ***lying over property** if every prime ideal $\mathfrak{p} \supseteq \ker(\phi)$ has a prime $\mathfrak{q}$ lying over it. NB $\ker(\phi) \subseteq \mathfrak{p}$ is a necessary condition for $\mathfrak{p}$ to be a contraction and is equivalent to $B_{\mathfrak{p}} \neq 0$.*
b) ***going up property** if for every pair of prime ideals $\mathfrak{p} \subsetneq \mathfrak{p}'$ in A and $\mathfrak{q} \lhd B$ lying over $\mathfrak{p}$, there exists a prime ideal $\mathfrak{q}'$ such that $\mathfrak{q} \subsetneq \mathfrak{q}'$ and $\mathfrak{q}'$ lies over $\mathfrak{p}'$.*
c) ***incomparability property** if for every pair of prime ideals $\mathfrak{q}, \mathfrak{q}' \lhd B$ then $\mathfrak{q} \subsetneq \mathfrak{q}' \implies \phi^{-1}(\mathfrak{q}) \subsetneq \phi^{-1}(\mathfrak{q}')$*
d) ***going down property** if for every pair of prime ideals $\mathfrak{p}' \subsetneq \mathfrak{p}$ in A and $\mathfrak{q} \lhd B$ lying over $\mathfrak{p}$, there exists a prime ideal $\mathfrak{q}'$ such that $\mathfrak{q}' \subsetneq \mathfrak{q}$ and $\mathfrak{q}'$ lies over $\mathfrak{p}'$.*

Remark 3.22.3
It's possible to interpret these geometrically in terms of the map $\phi_\star : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$.

- *Lying over - $\phi_\star$ is surjective onto $V(\ker(\phi))$*
- *Going up - $\phi_\star$ is closed*
- *Incomparability - fibres have dimension 0*
- *Going down (and finite presentation) - $\phi_\star$ is open*

The main result of this section is the correspondence between primes lying over $\mathfrak{p}$ and primes of the ring $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ (3.22.5). As a preliminary result we consider conditions under which we can strengthen $\mathfrak{p} \subseteq \mathfrak{q}^c$ to $\mathfrak{p} = \mathfrak{q}^c$, as the former condition is somewhat easier to satisfy.

Lemma 3.22.4 (Lying over criteria)
Let $\phi : A \to B$ be a ring map, $\mathfrak{p} \lhd A$ prime such that $\ker(\phi) \subseteq \mathfrak{p}$ and $\mathfrak{q} \lhd B$. Then

$$\mathfrak{p} = \mathfrak{q}^c \iff \mathfrak{p}^e \subseteq \mathfrak{q} \text{ and } \mathfrak{q} \cap \phi(A \setminus \mathfrak{p}) = \emptyset$$

In particular

$$\mathfrak{p} = \mathfrak{p}^{ec} \text{ is contracted} \iff \mathfrak{p}^e \cap \phi(A \setminus \mathfrak{p}) = \emptyset$$

Proof. Recall (3.3.52) that in general $\mathfrak{p} \subseteq \mathfrak{q}^c \iff \mathfrak{p}^e \subseteq \mathfrak{q}$. The first equivalence is then clear because $x \in \mathfrak{q}^c \setminus \mathfrak{p} \iff \phi(x) \in \mathfrak{q} \cap \phi(A \setminus \mathfrak{p})$. For $\phi(x) \in \mathfrak{q} \cap \phi(A \setminus \mathfrak{p}) \implies \phi(x) = \phi(y)$ for $y \notin \mathfrak{p}$ and $x \in \mathfrak{q}^c$. This implies $x - y \in \ker(\phi) \subseteq \mathfrak{p}$ whence $x \notin \mathfrak{p}$ as required.

The final statement follows by considering $\mathfrak{q} = \mathfrak{p}^e$. ∎

Proposition 3.22.5 (Lying over correspondence)
Let $\phi : A \to B$ be a ring map and $\mathfrak{p} \lhd A$ a prime ideal s.t. $\ker(\phi) \subseteq \mathfrak{p}$. Then there is a order-preserving correspondence of prime ideals

$$\begin{aligned} \{\mathfrak{q} \mid \mathfrak{q} \text{ lies above } \mathfrak{p}\} &\longleftrightarrow \left\{\mathfrak{q}' \lhd B_{\mathfrak{p}} \;\middle|\; \mathfrak{p}B_{\mathfrak{p}} \subseteq \mathfrak{q}'\right\} &&\longleftrightarrow \left\{\mathfrak{q}'' \lhd B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}\right\} \\ \mathfrak{q} &\longrightarrow \mathfrak{q}B_{\mathfrak{p}} && \longrightarrow \mathfrak{q}B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} \end{aligned}$$

In particular TFAE

a) *$\mathfrak{p}$ lies under a prime $\mathfrak{q}$*
b) *$B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} \neq 0$*
c) *$\mathfrak{p} = \mathfrak{p}^{ec}$ is contracted*

NB c) is a-priori weaker than a).

Proof. Recall $B_{\mathfrak{p}} := S^{-1}B$ and $\mathfrak{p}B_{\mathfrak{p}} := \mathfrak{p}^e B_{\mathfrak{p}} = S^{-1}\mathfrak{p}^e$ where $S := \phi(A \setminus \mathfrak{p})$.

By (3.22.4) $\mathfrak{q}$ lies above $\mathfrak{p}$ if and only if $\mathfrak{p}^e \subseteq \mathfrak{q}$ and $\mathfrak{q} \cap S = \emptyset$. The first correspondence then follows from (3.5.19) and the second from (3.3.56).

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a) $\iff$ b) This follows from the correspondence, since a ring without any non-zero prime ideals is simply the zero-ring.

b) $\iff$ c) Observe $\mathfrak{p} = \mathfrak{p}^{ec} \iff \mathfrak{p}^e \cap S = \emptyset$ by (3.22.4), which holds if and only if $\mathfrak{p}B_\mathfrak{p} \neq B_\mathfrak{p}$ by (3.5.18) e). ∎

Remark 3.22.6
Geometrically this is an explicit representation of the fiber as a prime spectrum

$$\operatorname{Spec}(\phi)^{-1}(\mathfrak{p}) \longleftrightarrow \operatorname{Spec}\big(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}\big)$$

Proposition 3.22.7 (Injective Ring Maps are Dense)
Let $\phi : A \to B$ be a ring map and $\mathfrak{p} \lhd A$ a minimal prime over $\ker(\phi)$. Then there exists a prime ideal $\mathfrak{q} \lhd B$ such that $\mathfrak{q}^c = \mathfrak{p}$. Furthermore $\mathfrak{q}$ may be taken to be minimal.

Proof. We may reduce to the case ϕ is injective by replacing A with $A/\ker(\phi)$. We have the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_T} \\ A_\mathfrak{p} & \overset{\tilde{\phi}}{\dashrightarrow} & T^{-1}B \end{array}$$

where $S := A \setminus \mathfrak{p}$ and $T := \phi(S)$. By (3.5.5) there exists a homomorphism $\tilde{\phi}$. Define $\mathfrak{m} := \mathfrak{p}A_\mathfrak{p}$ to be the unique maximal ideal of $A_\mathfrak{p}$. By (3.5.18) it is a minimal prime ideal and therefore the unique prime ideal. By assumption $0 \notin T$ and so $T^{-1}B \neq 0$. $T^{-1}B$ has a maximal ideal $\mathfrak{n}$ by (3.3.37). By uniqueness we find $\tilde{\phi}^{-1}(\mathfrak{n}) = \mathfrak{m}$. Therefore $\mathfrak{q} := i_T^{-1}(\mathfrak{n})$ is a prime ideal such that $\mathfrak{q}^c = \mathfrak{p}$.

By (3.3.44) there is a minimal prime $\mathfrak{r} \subset \mathfrak{q}$ which by minimality also satisfies $\mathfrak{r}^c = \mathfrak{p}$. ∎

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